Install or upgrade the RCF

Install and maintain

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Install or upgrade the RCF

Install or upgrade the Reference Configuration File (RCF) overview

You install the Reference Configuration File (RCF) after setting up the Nexus 3132Q-V switches for the first time. You upgrade your RCF version when you have an existing version of the RCF file installed on your switch.

Refer to the Knowledge Base article How to clear configuration on a Cisco interconnect switch while retaining remote connectivity for further information when installing or upgrading your RCF.

Available RCF configurations

The following table describes the RCFs available for different configurations. Choose the RCF applicable to your configuration.

For specific port and VLAN usage details, refer to the banner and important notes section in your RCF.

RCF name	Description
2-Cluster-HA-Breakout	Supports two ONTAP clusters with at least eight nodes, including nodes that use shared Cluster+HA ports.
4-Cluster-HA-Breakout	Supports four ONTAP clusters with at least four nodes, including nodes that use shared Cluster+HA ports.
1-Cluster-HA	All ports are configured for 40/100GbE. Supports shared cluster/HA traffic on ports. Required for AFF A320, AFF A250, and FAS500f systems. Additionally, all ports can be used as dedicated cluster ports.
1-Cluster-HA-Breakout	Ports are configured for 4x10GbE breakout, 4x25GbE breakout (RCF 1.6+ on 100GbE switches), and 40/100GbE. Supports shared cluster/HA traffic on ports for nodes that use shared cluster/HA ports: AFF A320, AFF A250, and FAS500f systems. Additionally, all ports can be used as dedicated cluster ports.
Cluster-HA-Storage	Ports are configured for 40/100GbE for Cluster+HA, 4x10GbE breakout for Cluster and 4x25GbE breakout for Cluster+HA, and 100GbE for each Storage HA Pair.
Cluster	Two flavors of RCF with different allocations of 4x10GbE ports (breakout) and 40/100GbE ports. All FAS/AFF nodes are supported, except for AFF A320, AFF A250, and FAS500f systems.
Storage	All ports are configured for 100GbE NVMe storage connections.

Available RCFs

The following table lists the available RCFs for 3132Q-V switches. Choose the applicable RCF version for your configuration. Refer to Cisco Ethernet Switches for more information.

RCF name
Cluster-HA-Breakout RCF v1.*xx*
Cluster-HA RCF v1.*xx*
Cluster RCF 1.*xx*

Suggested documentation

- Cisco Ethernet Switches (NSS)

  Consult the switch compatibility table for the supported ONTAP and RCF versions on the NetApp Support Site. Note that there can be command dependencies between the command syntax in the RCF and the syntax found in specific versions of NX-OS.

- Cisco Nexus 3000 Series Switches

  Refer to the appropriate software and upgrade guides available on the Cisco website for complete documentation on the Cisco switch upgrade and downgrade procedures.

About the examples

The examples in this procedure use the following switch and node nomenclature:

- The names of the two Cisco switches are **cs1** and **cs2**.
- The node names are **cluster1-01**, **cluster1-02**, **cluster1-03**, and **cluster1-04**.
- The cluster LIF names are **cluster1-01_clus1**, **cluster1-01_clus2**, **cluster1-02_clus1**, **cluster1-02_clus2**, **cluster1-03_clus1**, **cluster1-03_clus2**, **cluster1-04_clus1**, and **cluster1-04_clus2**.
- The `cluster1::*>` prompt indicates the name of the cluster.

The examples in this procedure use four nodes. These nodes use two 10GbE cluster interconnect ports **e0a** and **e0b**. Refer to the Hardware Universe to verify the correct cluster ports on your platforms.

> ⓘ The command outputs might vary depending on different releases of ONTAP.

For details of the available RCF configurations, refer to Software install workflow.

Commands used

The procedure requires the use of both ONTAP commands and Cisco Nexus 3000 Series Switches commands; ONTAP commands are used unless otherwise indicated.

What's next?

After you've reviewed the install RCF or upgrade RCF procedure, you install the RCF or upgrade your RCF as required.

Install the Reference Configuration File (RCF)

You install the Reference Configuration File (RCF) after setting up the Nexus 3132Q-V switches for the first time.

Before you begin

Verify the following installations and connections:

- A current backup of the switch configuration.
- A fully functioning cluster (no errors in the logs or similar issues).
- The current RCF.
- A console connection to the switch, required when installing the RCF.

About this task

The procedure requires the use of both ONTAP commands and Cisco Nexus 3000 Series Switches commands; ONTAP commands are used unless otherwise indicated.

No operational inter-switch link (ISL) is needed during this procedure. This is by design because RCF version changes can affect ISL connectivity temporarily. To enable non-disruptive cluster operations, the following procedure migrates all of the cluster LIFs to the operational partner switch while performing the steps on the target switch.

Step 1: Install the RCF on the switches

1. Display the cluster ports on each node that are connected to the cluster switches:

   ```
   network device-discovery show
   ```

Show example

```
cluster1::*> network device-discovery show
Node/       Local  Discovered
Protocol    Port   Device (LLDP: ChassisID)  Interface
Platform
----------- ------ ------------------------- ----------------
------------
cluster1-01/cdp
            e0a    cs1                       Ethernet1/7      N3K-
C3132Q-V
            e0d    cs2                       Ethernet1/7      N3K-
C3132Q-V
cluster1-02/cdp
            e0a    cs1                       Ethernet1/8      N3K-
C3132Q-V
            e0d    cs2                       Ethernet1/8      N3K-
C3132Q-V
cluster1-03/cdp
            e0a    cs1                       Ethernet1/1/1    N3K-
C3132Q-V
            e0b    cs2                       Ethernet1/1/1    N3K-
C3132Q-V
cluster1-04/cdp
            e0a    cs1                       Ethernet1/1/2    N3K-
C3132Q-V
            e0b    cs2                       Ethernet1/1/2    N3K-
C3132Q-V
cluster1::*>
```

2. Check the administrative and operational status of each cluster port.

   a. Verify that all the cluster ports are up with a healthy status:

      `network port show -ipspace Cluster`

Show example

```
cluster1::*> network port show -ipspace Cluster
Node: cluster1-01

Ignore
                                                  Speed(Mbps)
Health   Health
Port      IPspace      Broadcast Domain Link MTU  Admin/Oper
Status   Status
--------- ------------ ---------------- ---- ---- -----------
-------- ------
e0a       Cluster      Cluster          up   9000  auto/100000
healthy false
e0d       Cluster      Cluster          up   9000  auto/100000
healthy false
Node: cluster1-02

Ignore
                                                  Speed(Mbps)
Health   Health
Port      IPspace      Broadcast Domain Link MTU  Admin/Oper
Status   Status
--------- ------------ ---------------- ---- ---- -----------
-------- ------
e0a       Cluster      Cluster          up   9000  auto/100000
healthy false
e0d       Cluster      Cluster          up   9000  auto/100000
healthy false
8 entries were displayed.
Node: cluster1-03
   Ignore
                                                  Speed(Mbps)
Health   Health
Port      IPspace      Broadcast Domain Link MTU  Admin/Oper
Status   Status
--------- ------------ ---------------- ---- ---- -----------
-------- ------
e0a       Cluster      Cluster          up   9000  auto/10000
healthy  false
e0b       Cluster      Cluster          up   9000  auto/10000
healthy  false
Node: cluster1-04

Ignore
                                                  Speed(Mbps)
```

```
Health   Health
Port      IPspace      Broadcast Domain Link MTU  Admin/Oper
Status   Status
--------- ------------ ---------------- ---- ---- -----------
-------- ------
e0a       Cluster      Cluster          up   9000  auto/10000
healthy  false
e0b       Cluster      Cluster          up   9000  auto/10000
healthy  false
cluster1::*>
```

b. Verify that all the cluster interfaces (LIFs) are on the home port:

`network interface show -vserver Cluster`

Show example

```
cluster1::*> network interface show -vserver Cluster
            Logical            Status     Network
Current      Current Is
Vserver     Interface          Admin/Oper Address/Mask      Node
Port    Home
----------- ------------------ ---------- -----------------
------------ ------- ----
Cluster
            cluster1-01_clus1  up/up     169.254.3.4/23
cluster1-01  e0a     true
            cluster1-01_clus2  up/up     169.254.3.5/23
cluster1-01  e0d     true
            cluster1-02_clus1  up/up     169.254.3.8/23
cluster1-02  e0a     true
            cluster1-02_clus2  up/up     169.254.3.9/23
cluster1-02  e0d     true
            cluster1-03_clus1  up/up     169.254.1.3/23
cluster1-03  e0a     true
            cluster1-03_clus2  up/up     169.254.1.1/23
cluster1-03  e0b     true
            cluster1-04_clus1  up/up     169.254.1.6/23
cluster1-04  e0a     true
            cluster1-04_clus2  up/up     169.254.1.7/23
cluster1-04  e0b     true
cluster1::*>
```

c. Verify that the cluster displays information for both cluster switches:

`system cluster-switch show -is-monitoring-enabled-operational true`

Show example

```
cluster1::*> system cluster-switch show -is-monitoring-enabled
-operational true
Switch                      Type               Address
Model
--------------------------- ------------------ ----------------
---------------
cs1                         cluster-network    10.0.0.1
NX3132QV
     Serial Number: FOXXXXXXXGS
      Is Monitored: true
            Reason: None
  Software Version: Cisco Nexus Operating System (NX-OS)
Software, Version
                    9.3(4)
    Version Source: CDP
cs2                         cluster-network    10.0.0.2
NX3132QV
     Serial Number: FOXXXXXXXGD
      Is Monitored: true
            Reason: None
  Software Version: Cisco Nexus Operating System (NX-OS)
Software, Version
                    9.3(4)
    Version Source: CDP
2 entries were displayed.
```

ⓘ For ONTAP 9.8 and later, use the command `system switch ethernet show -is -monitoring-enabled-operational true`.

3. Disable auto-revert on the cluster LIFs.

```
cluster1::*> network interface modify -vserver Cluster -lif * -auto
-revert false
```

Make sure that auto-revert is disabled after running this command.

4. On cluster switch cs2, shut down the ports connected to the cluster ports of the nodes.

```
cs2> enable
cs2# configure
cs2(config)# interface eth1/1/1-2,eth1/7-8
cs2(config-if-range)# shutdown
cs2(config-if-range)# exit
cs2# exit
```

The number of ports displayed varies based on the number of nodes in the cluster.

5. Verify that the cluster ports have failed over to the ports hosted on cluster switch cs1. This might take a few seconds.

   `network interface show -vserver Cluster`

   Show example

```
cluster1::*> network interface show -vserver Cluster
            Logical           Status     Network            Current
Current Is
Vserver     Interface         Admin/Oper Address/Mask       Node
Port    Home
----------- ----------------- ---------- ------------------
------------- ------- ----
Cluster
            cluster1-01_clus1 up/up      169.254.3.4/23
cluster1-01   e0a     true
            cluster1-01_clus2 up/up      169.254.3.5/23
cluster1-01   e0a     false
            cluster1-02_clus1 up/up      169.254.3.8/23
cluster1-02   e0a     true
            cluster1-02_clus2 up/up      169.254.3.9/23
cluster1-02   e0a     false
            cluster1-03_clus1 up/up      169.254.1.3/23
cluster1-03   e0a     true
            cluster1-03_clus2 up/up      169.254.1.1/23
cluster1-03   e0a     false
            cluster1-04_clus1 up/up      169.254.1.6/23
cluster1-04   e0a     true
            cluster1-04_clus2 up/up      169.254.1.7/23
cluster1-04   e0a     false
cluster1::*>
```

6. Verify that the cluster is healthy:

```
cluster show
```

Show example

```
cluster1::*> cluster show
Node                 Health  Eligibility   Epsilon
-------------------- ------- ------------  -------
cluster1-01          true    true          false
cluster1-02          true    true          false
cluster1-03          true    true          true
cluster1-04          true    true          false
cluster1::*>
```

7. If you have not already done so, save a copy of the current switch configuration by copying the output of the following command to a text file:

   ```
   show running-config
   ```

8. Record any custom additions between the current running-config and the RCF file in use.

   > Make sure to configure the following: * Username and password * Management IP address * Default gateway * Switch name

9. Save basic configuration details to the `write_erase.cfg` file on the bootflash.

   > When upgrading or applying a new RCF, you must erase the switch settings and perform basic configuration. You must be connected to the switch serial console port to set up the switch again.

   ```
   cs2# show run | section "switchname" > bootflash:write_erase.cfg
   ```

   ```
   cs2# show run | section "hostname" >> bootflash:write_erase.cfg
   ```

   ```
   cs2# show run | i "username admin password" >> bootflash:write_erase.cfg
   ```

   ```
   cs2# show run | section "vrf context management" >> bootflash:write_erase.cfg
   ```

   ```
   cs2# show run | section "interface mgmt0" >> bootflash:write_erase.cfg
   ```

10. When installing RCF version 1.12 and later, run the following commands:

    ```
    cs2# echo "hardware access-list tcam region vpc-convergence 256" >>
    bootflash:write_erase.cfg
    ```

    ```
    cs2# echo "hardware access-list tcam region racl 256" >>
    bootflash:write_erase.cfg
    ```

    ```
    cs2# echo "hardware access-list tcam region e-racl 256" >>
    ```

```
bootflash:write_erase.cfg

cs2# echo "hardware access-list tcam region qos 256" >>
bootflash:write_erase.cfg
```

Refer to How to clear configuration on a Cisco interconnect switch while retaining remote connectivity for further details.

11. Verify that the `write_erase.cfg` file is populated as expected:

    ```
    show file bootflash:write_erase.cfg
    ```

12. Issue the `write erase` command to erase the current saved configuration:

    ```
    cs2# write erase

    Warning: This command will erase the startup-configuration.

    Do you wish to proceed anyway? (y/n)  [n] y
    ```

13. Copy the previously saved basic configuration into the startup configuration.

    ```
    cs2# copy bootflash:write_erase.cfg startup-config
    ```

14. Reboot the switch:

    ```
    cs2# reload

    This command will reboot the system. (y/n)?  [n] y
    ```

15. Repeat Steps 7 to 14 on switch cs1.
16. Connect the cluster ports of all nodes in the ONTAP cluster to switches cs1 and cs2.

Step 2: Verify the switch connections

1. Verify that the switch ports connected to the cluster ports are **up**.

   ```
   show interface brief | grep up
   ```

Show example

```
cs1# show interface brief | grep up
.
.
Eth1/1/1      1       eth  access up      none
10G(D) --
Eth1/1/2      1       eth  access up      none
10G(D) --
Eth1/7        1       eth  trunk  up      none
100G(D) --
Eth1/8        1       eth  trunk  up      none
100G(D) --
.
.
```

2. Verify that the ISL between cs1 and cs2 is functional:

`show port-channel summary`

Show example

```
cs1# show port-channel summary
Flags:  D - Down        P - Up in port-channel (members)
        I - Individual  H - Hot-standby (LACP only)
        s - Suspended   r - Module-removed
        b - BFD Session Wait
        S - Switched    R - Routed
        U - Up (port-channel)
        p - Up in delay-lacp mode (member)
        M - Not in use. Min-links not met
--------------------------------------------------------------------
------------
Group Port-       Type     Protocol  Member Ports
      Channel
--------------------------------------------------------------------
------------
1     Po1(SU)     Eth      LACP      Eth1/31(P)   Eth1/32(P)
cs1#
```

3. Verify that the cluster LIFs have reverted to their home port:

`network interface show -vserver Cluster`

Show example

```
cluster1::*> network interface show -vserver Cluster
            Logical            Status     Network            Current
Current Is
Vserver     Interface          Admin/Oper Address/Mask       Node
Port    Home
----------- ------------------ ---------- ------------------
------------------- ------- ----
Cluster
            cluster1-01_clus1  up/up      169.254.3.4/23
cluster1-01         e0d     true
            cluster1-01_clus2  up/up      169.254.3.5/23
cluster1-01         e0d     true
            cluster1-02_clus1  up/up      169.254.3.8/23
cluster1-02         e0d     true
            cluster1-02_clus2  up/up      169.254.3.9/23
cluster1-02         e0d     true
            cluster1-03_clus1  up/up      169.254.1.3/23
cluster1-03         e0b     true
            cluster1-03_clus2  up/up      169.254.1.1/23
cluster1-03         e0b     true
            cluster1-04_clus1  up/up      169.254.1.6/23
cluster1-04         e0b     true
            cluster1-04_clus2  up/up      169.254.1.7/23
cluster1-04         e0b     true
cluster1::*>
```

4. Verify that the cluster is healthy:

`cluster show`

Show example

```
cluster1::*> cluster show
Node                 Health  Eligibility   Epsilon
-------------------- ------- ------------- -------
cluster1-01          true    true          false
cluster1-02          true    true          false
cluster1-03          true    true          true
cluster1-04          true    true          false
cluster1::*>
```

Step 3: Setup your ONTAP cluster

NetApp recommends that you use System Manager to set up new clusters.

System Manager provides a simple and easy workflow for cluster set up and configuration including assigning a node management IP address, initializing the cluster, creating a local tier, configuring protocols, and provisioning initial storage.

Refer to Configure ONTAP on a new cluster with System Manager for setup instructions.

What's next?

After you've installed the RCF, you can verify the SSH configuration.

Upgrade your Reference Configuration File (RCF)

You upgrade your RCF version when you have an existing version of the RCF file installed on your operational switches.

Before you begin

Make sure you have the following:

- A current backup of the switch configuration.
- A fully functioning cluster (no errors in the logs or similar issues).
- The current RCF.
- If you are updating your RCF version, you need a boot configuration in the RCF that reflects the desired boot images.

  If you need to change the boot configuration to reflect the current boot images, you must do so before reapplying the RCF so that the correct version is instantiated on future reboots.

> No operational inter-switch link (ISL) is needed during this procedure. This is by design because RCF version changes can affect ISL connectivity temporarily. To ensure non-disruptive cluster operations, the following procedure migrates all of the cluster LIFs to the operational partner switch while performing the steps on the target switch.

> Before installing a new switch software version and RCFs, you must erase the switch settings and perform basic configuration. You must be connected to the switch using the serial console or have preserved basic configuration information prior to erasing the switch settings.

Step 1: Prepare for the upgrade

1. Display the cluster ports on each node that are connected to the cluster switches:

   ```
   network device-discovery show
   ```

Show example

```
cluster1::*> network device-discovery show
Node/        Local  Discovered
Protocol     Port   Device (LLDP: ChassisID)  Interface
Platform
----------- ------ ------------------------- ----------------
------------
cluster1-01/cdp
             e0a    cs1                       Ethernet1/7       N3K-
C3132Q-V
             e0d    cs2                       Ethernet1/7       N3K-
C3132Q-V
cluster1-02/cdp
             e0a    cs1                       Ethernet1/8       N3K-
C3132Q-V
             e0d    cs2                       Ethernet1/8       N3K-
C3132Q-V
cluster1-03/cdp
             e0a    cs1                       Ethernet1/1/1     N3K-
C3132Q-V
             e0b    cs2                       Ethernet1/1/1     N3K-
C3132Q-V
cluster1-04/cdp
             e0a    cs1                       Ethernet1/1/2     N3K-
C3132Q-V
             e0b    cs2                       Ethernet1/1/2     N3K-
C3132Q-V
cluster1::*>
```

2. Check the administrative and operational status of each cluster port.

   a. Verify that all the cluster ports are up with a healthy status:

      `network port show -ipspace Cluster`

Show example

```
cluster1::*> network port show -ipspace Cluster

Node: cluster1-01

Ignore
                                                  Speed(Mbps)
Health   Health
Port      IPspace      Broadcast Domain Link MTU  Admin/Oper
Status   Status
--------- ------------ ---------------- ---- ---- -----------
-------- ------
e0a       Cluster      Cluster          up   9000  auto/100000
healthy false
e0d       Cluster      Cluster          up   9000  auto/100000
healthy false

Node: cluster1-02

Ignore
                                                  Speed(Mbps)
Health   Health
Port      IPspace      Broadcast Domain Link MTU  Admin/Oper
Status   Status
--------- ------------ ---------------- ---- ---- -----------
-------- ------
e0a       Cluster      Cluster          up   9000  auto/100000
healthy false
e0d       Cluster      Cluster          up   9000  auto/100000
healthy false
8 entries were displayed.

Node: cluster1-03

   Ignore
                                                  Speed(Mbps)
Health   Health
Port      IPspace      Broadcast Domain Link MTU  Admin/Oper
Status   Status
--------- ------------ ---------------- ---- ---- -----------
-------- ------
e0a       Cluster      Cluster          up   9000  auto/10000
healthy  false
e0b       Cluster      Cluster          up   9000  auto/10000
healthy  false
```

```
Node: cluster1-04

Ignore
                                                     Speed(Mbps)
Health   Health
Port      IPspace      Broadcast Domain Link MTU  Admin/Oper
Status   Status
--------- ------------ ---------------- ---- ---- -----------
-------- ------
e0a       Cluster      Cluster          up   9000  auto/10000
healthy  false
e0b       Cluster      Cluster          up   9000  auto/10000
healthy  false
cluster1::*>
```

b. Verify that all the cluster interfaces (LIFs) are on the home port:

```
network interface show -vserver Cluster
```

Show example

```
cluster1::*> network interface show -vserver Cluster
            Logical            Status     Network
Current      Current Is
Vserver     Interface          Admin/Oper Address/Mask      Node
Port    Home
----------- ------------------ ---------- -----------------
------------ ------- ----
Cluster
            cluster1-01_clus1  up/up     169.254.3.4/23
cluster1-01  e0a     true
            cluster1-01_clus2  up/up     169.254.3.5/23
cluster1-01  e0d     true
            cluster1-02_clus1  up/up     169.254.3.8/23
cluster1-02  e0a     true
            cluster1-02_clus2  up/up     169.254.3.9/23
cluster1-02  e0d     true
            cluster1-03_clus1  up/up     169.254.1.3/23
cluster1-03  e0a     true
            cluster1-03_clus2  up/up     169.254.1.1/23
cluster1-03  e0b     true
            cluster1-04_clus1  up/up     169.254.1.6/23
cluster1-04  e0a     true
            cluster1-04_clus2  up/up     169.254.1.7/23
cluster1-04  e0b     true
cluster1::*>
```

c. Verify that the cluster displays information for both cluster switches:

`system cluster-switch show -is-monitoring-enabled-operational true`

Show example

```
cluster1::*> system cluster-switch show -is-monitoring-enabled
-operational true
Switch                           Type               Address
Model
--------------------------- ------------------ ----------------
---------------
cs1                              cluster-network    10.0.0.1
NX3132QV
      Serial Number: FOXXXXXXXGS
       Is Monitored: true
             Reason: None
   Software Version: Cisco Nexus Operating System (NX-OS)
Software, Version
                     9.3(4)
     Version Source: CDP

cs2                              cluster-network    10.0.0.2
NX3132QV
      Serial Number: FOXXXXXXXGD
       Is Monitored: true
             Reason: None
   Software Version: Cisco Nexus Operating System (NX-OS)
Software, Version
                     9.3(4)
     Version Source: CDP

2 entries were displayed.
```

For ONTAP 9.8 and later, use the command `system switch ethernet show -is -monitoring-enabled-operational true`.

3. Disable auto-revert on the cluster LIFs.

```
cluster1::*> network interface modify -vserver Cluster -lif * -auto
-revert false
```

Make sure that auto-revert is disabled after running this command.

Step 2: Configure ports

1. On cluster switch cs2, shut down the ports connected to the cluster ports of the nodes.

```
cs2> enable
cs2# configure
cs2(config)# interface eth1/1/1-2,eth1/7-8
cs2(config-if-range)# shutdown
cs2(config-if-range)# exit
cs2# exit
```

The number of ports displayed varies based on the number of nodes in the cluster.

2. Verify that the cluster ports have failed over to the ports hosted on cluster switch cs1. This might take a few seconds.

`network interface show -vserver Cluster`

Show example

```
cluster1::*> network interface show -vserver Cluster
            Logical           Status     Network            Current
Current Is
Vserver     Interface         Admin/Oper Address/Mask       Node
Port    Home
----------- ----------------- ---------- ------------------
------------- ------- ----
Cluster
            cluster1-01_clus1 up/up      169.254.3.4/23
cluster1-01   e0a     true
            cluster1-01_clus2 up/up      169.254.3.5/23
cluster1-01   e0a     false
            cluster1-02_clus1 up/up      169.254.3.8/23
cluster1-02   e0a     true
            cluster1-02_clus2 up/up      169.254.3.9/23
cluster1-02   e0a     false
            cluster1-03_clus1 up/up      169.254.1.3/23
cluster1-03   e0a     true
            cluster1-03_clus2 up/up      169.254.1.1/23
cluster1-03   e0a     false
            cluster1-04_clus1 up/up      169.254.1.6/23
cluster1-04   e0a     true
            cluster1-04_clus2 up/up      169.254.1.7/23
cluster1-04   e0a     false
cluster1::*>
```

3. Verify that the cluster is healthy:

```
cluster show
```

Show example

```
cluster1::*> cluster show
Node                 Health  Eligibility   Epsilon
-------------------- ------- ------------  -------
cluster1-01          true    true          false
cluster1-02          true    true          false
cluster1-03          true    true          true
cluster1-04          true    true          false
cluster1::*>
```

4. If you have not already done so, save a copy of the current switch configuration by copying the output of the following command to a text file:

```
show running-config
```

5. Record any custom additions between the current running-config and the RCF file in use.

> Make sure to configure the following:
>
> - Username and password
> - Management IP address
> - Default gateway
> - Switch name

6. Save basic configuration details to the `write_erase.cfg` file on the bootflash.

> When upgrading or applying a new RCF, you must erase the switch settings and perform basic configuration.

```
cs2# show run | section "switchname" > bootflash:write_erase.cfg
```

```
cs2# show run | section "hostname" >> bootflash:write_erase.cfg
```

```
cs2# show run | i "username admin password" >> bootflash:write_erase.cfg
```

```
cs2# show run | section "vrf context management" >> bootflash:write_erase.cfg
```

```
cs2# show run | section "interface mgmt0" >> bootflash:write_erase.cfg
```

7. When upgrading to RCF version 1.12 and later, run the following commands:

```
cs2# echo "hardware access-list tcam region vpc-convergence 256" >>
bootflash:write_erase.cfg
```

```
cs2# echo "hardware access-list tcam region racl 256" >>
bootflash:write_erase.cfg

cs2# echo "hardware access-list tcam region e-racl 256" >>
bootflash:write_erase.cfg

cs2# echo "hardware access-list tcam region qos 256" >>
bootflash:write_erase.cfg
```

8. Verify that the `write_erase.cfg` file is populated as expected:

   ```
   show file bootflash:write_erase.cfg
   ```

9. Issue the `write erase` command to erase the current saved configuration:

   ```
   cs2# write erase

   Warning: This command will erase the startup-configuration.

   Do you wish to proceed anyway? (y/n) [n] y
   ```

10. Copy the previously saved basic configuration into the startup configuration.

    ```
    cs2# copy bootflash:write_erase.cfg startup-config
    ```

11. Reboot the switch:

    ```
    cs2# reload

    This command will reboot the system. (y/n)? [n] y
    ```

12. After the management IP address is reachable again, log in to the switch through SSH.

    You may need to update host file entries related to the SSH keys.

13. Copy the RCF to the bootflash of switch cs2 using one of the following transfer protocols: FTP, TFTP, SFTP, or SCP. For more information on Cisco commands, refer to the appropriate guide in the Cisco Nexus 3000 Series NX-OS Command Reference.

    Show example

    ```
    cs2# copy tftp: bootflash: vrf management
    Enter source filename: Nexus_3132QV_RCF_v1.6-Cluster-HA-Breakout.txt
    Enter hostname for the tftp server: 172.22.201.50
    Trying to connect to tftp server......Connection to Server
    Established.
    TFTP get operation was successful
    Copy complete, now saving to disk (please wait)...
    ```

14. Apply the RCF previously downloaded to the bootflash.

    For more information on Cisco commands, refer to the appropriate guide in the Cisco Nexus 3000 Series NX-OS Command Reference.

    Show example

    ```
    cs2# copy Nexus_3132QV_RCF_v1.6-Cluster-HA-Breakout.txt running-
    config echo-commands
    ```

    > Make sure to read thoroughly the **Installation notes**, **Important Notes**, and **banner** sections of your RCF. You must read and follow these instructions to ensure the proper configuration and operation of the switch.

15. Verify that the RCF file is the correct newer version:

    `show running-config`

    When you check the output to verify you have the correct RCF, make sure that the following information is correct:

    - The RCF banner
    - The node and port settings
    - Customizations

      The output varies according to your site configuration. Check the port settings and refer to the release notes for any changes specific to the RCF that you have installed.

      > For steps on how to bring your 10GbE ports online after an upgrade of the RCF, refer to the Knowledge Base article 10GbE ports on a Cisco 3132Q cluster switch do not come online.

16. After you verify the RCF versions and switch settings are correct, copy the `running-config` file to the `startup-config` file.

    For more information on Cisco commands, refer to the appropriate guide in the Cisco Nexus 3000 Series NX-OS Command Reference. .Show example

```
cs2# copy running-config startup-config
[########################################] 100% Copy complete
```

1. Reboot switch cs2. You can ignore both the "cluster ports down" events reported on the nodes while the switch reboots and the error `% Invalid command at '^' marker` output.

```
cs2# reload
This command will reboot the system. (y/n)?  [n] y
```

2. Reapply any previous customizations to the switch configuration. Refer to Review cabling and configuration considerations for details of any further changes required.
3. Verify the health of cluster ports on the cluster.
   a. Verify that cluster ports are up and healthy across all nodes in the cluster:

      ```
      network port show -ipspace Cluster
      ```

Show example

```
cluster1::*> network port show -ipspace Cluster

Node: cluster1-01

Ignore
                                                  Speed(Mbps)
Health   Health
Port      IPspace      Broadcast Domain Link MTU  Admin/Oper
Status   Status
--------- ------------ ---------------- ---- ---- -----------
-------- ------
e0a       Cluster      Cluster          up   9000  auto/10000
healthy  false
e0b       Cluster      Cluster          up   9000  auto/10000
healthy  false

Node: cluster1-02

Ignore
                                                  Speed(Mbps)
Health   Health
Port      IPspace      Broadcast Domain Link MTU  Admin/Oper
Status   Status
--------- ------------ ---------------- ---- ---- -----------
-------- ------
e0a       Cluster      Cluster          up   9000  auto/10000
healthy  false
e0b       Cluster      Cluster          up   9000  auto/10000
healthy  false

Node: cluster1-03

Ignore
                                                  Speed(Mbps)
Health   Health
Port      IPspace      Broadcast Domain Link MTU  Admin/Oper
Status   Status
--------- ------------ ---------------- ---- ---- -----------
-------- ------
e0a       Cluster      Cluster          up   9000  auto/100000
healthy false
e0d       Cluster      Cluster          up   9000  auto/100000
healthy false
```

```
Node: cluster1-04

Ignore
                                                  Speed(Mbps)
Health   Health
Port      IPspace      Broadcast Domain Link MTU  Admin/Oper
Status   Status
--------- ------------ ---------------- ---- ---- -----------
-------- ------
e0a       Cluster      Cluster          up   9000  auto/100000
healthy false
e0d       Cluster      Cluster          up   9000  auto/100000
healthy false
```

b. Verify the switch health from the cluster.

```
network device-discovery show -protocol cdp
```

Show example

```
cluster1::*> network device-discovery show -protocol cdp
Node/       Local  Discovered
Protocol    Port   Device (LLDP: ChassisID)  Interface
Platform
----------- ------ ------------------------- -----------------
--------
cluster1-01/cdp
            e0a    cs1                       Ethernet1/7
N3K-C3132Q-V
            e0d    cs2                       Ethernet1/7
N3K-C3132Q-V
cluster01-2/cdp
            e0a    cs1                       Ethernet1/8
N3K-C3132Q-V
            e0d    cs2                       Ethernet1/8
N3K-C3132Q-V
cluster01-3/cdp
            e0a    cs1                       Ethernet1/1/1
N3K-C3132Q-V
            e0b    cs2                       Ethernet1/1/1
N3K-C3132Q-V
cluster1-04/cdp
            e0a    cs1                       Ethernet1/1/2
N3K-C3132Q-V
            e0b    cs2                       Ethernet1/1/2
N3K-C3132Q-V

cluster1::*> system cluster-switch show -is-monitoring-enabled
-operational true
Switch                      Type               Address
Model
--------------------------- ------------------ ----------------
-----
cs1                         cluster-network    10.233.205.90
N3K-C3132Q-V
     Serial Number: FOXXXXXXXGD
      Is Monitored: true
            Reason: None
  Software Version: Cisco Nexus Operating System (NX-OS)
Software, Version
                    9.3(4)
    Version Source: CDP

cs2                         cluster-network    10.233.205.91
```

```
N3K-C3132Q-V
      Serial Number: FOXXXXXXXGS
       Is Monitored: true
             Reason: None
   Software Version: Cisco Nexus Operating System (NX-OS)
Software, Version
                     9.3(4)
     Version Source: CDP

2 entries were displayed.
```

For ONTAP 9.8 and later, use the command `system switch ethernet show -is -monitoring-enabled-operational true`.

You might observe the following output on the cs1 switch console depending on the RCF version previously loaded on the switch:

```
2020 Nov 17 16:07:18 cs1 %$ VDC-1 %$ %STP-2-
UNBLOCK_CONSIST_PORT: Unblocking port port-channel1 on
VLAN0092. Port consistency restored.
2020 Nov 17 16:07:23 cs1 %$ VDC-1 %$ %STP-2-
BLOCK_PVID_PEER: Blocking port-channel1 on VLAN0001.
Inconsistent peer vlan.
2020 Nov 17 16:07:23 cs1 %$ VDC-1 %$ %STP-2-
BLOCK_PVID_LOCAL: Blocking port-channel1 on VLAN0092.
Inconsistent local vlan.
```

It can take up to 5 minutes for the cluster nodes to report as healthy.

4. On cluster switch cs1, shut down the ports connected to the cluster ports of the nodes.

Show example

```
cs1> enable
cs1# configure
cs1(config)# interface eth1/1/1-2,eth1/7-8
cs1(config-if-range)# shutdown
cs1(config-if-range)# exit
cs1# exit
```

The number of ports displayed varies based on the number of nodes in the cluster.

5. Verify that the cluster LIFs have migrated to the ports hosted on switch cs2. This might take a few seconds.

   `network interface show -vserver Cluster`

   Show example

```
cluster1::*> network interface show -vserver Cluster
             Logical            Status     Network            Current
Current Is
Vserver      Interface          Admin/Oper Address/Mask       Node
Port    Home
----------- ------------------ ---------- ------------------
------------------- ------- ----
Cluster
             cluster1-01_clus1  up/up      169.254.3.4/23
cluster1-01         e0d     false
             cluster1-01_clus2  up/up      169.254.3.5/23
cluster1-01         e0d     true
             cluster1-02_clus1  up/up      169.254.3.8/23
cluster1-02         e0d     false
             cluster1-02_clus2  up/up      169.254.3.9/23
cluster1-02         e0d     true
             cluster1-03_clus1  up/up      169.254.1.3/23
cluster1-03         e0b     false
             cluster1-03_clus2  up/up      169.254.1.1/23
cluster1-03         e0b     true
             cluster1-04_clus1  up/up      169.254.1.6/23
cluster1-04         e0b     false
             cluster1-04_clus2  up/up      169.254.1.7/23
cluster1-04         e0b     true
cluster1::*>
```

6. Verify that the cluster is healthy:

   `cluster show`

Show example

```
cluster1::*> cluster show
Node                 Health   Eligibility   Epsilon
-------------------- -------- ------------- -------
cluster1-01          true     true          false
cluster1-02          true     true          false
cluster1-03          true     true          true
cluster1-04          true     true          false
4 entries were displayed.
cluster1::*>
```

7. Repeat Steps 1 to 19 on switch cs1.
8. Enable auto-revert on the cluster LIFs.

```
cluster1::*> network interface modify -vserver Cluster -lif * -auto
-revert True
```

9. Reboot switch cs1. You do this to trigger the cluster LIFs to revert to their home ports. You can ignore the "cluster ports down" events reported on the nodes while the switch reboots.

```
cs1# reload
This command will reboot the system. (y/n)?  [n] y
```

Step 3: Verify the configuration

1. Verify that the switch ports connected to the cluster ports are up.

```
show interface brief | grep up
```

Show example

```
cs1# show interface brief | grep up
.
.
Eth1/1/1       1       eth  access up      none
10G(D) --
Eth1/1/2       1       eth  access up      none
10G(D) --
Eth1/7         1       eth  trunk  up      none
100G(D) --
Eth1/8         1       eth  trunk  up      none
100G(D) --
.
.
```

2. Verify that the ISL between cs1 and cs2 is functional:

`show port-channel summary`

Show example

```
cs1# show port-channel summary
Flags:  D - Down         P - Up in port-channel (members)
        I - Individual  H - Hot-standby (LACP only)
        s - Suspended   r - Module-removed
        b - BFD Session Wait
        S - Switched    R - Routed
        U - Up (port-channel)
        p - Up in delay-lacp mode (member)
        M - Not in use. Min-links not met
--------------------------------------------------------------------
------------
Group Port-       Type     Protocol  Member Ports
      Channel
--------------------------------------------------------------------
------------
1     Po1(SU)     Eth      LACP      Eth1/31(P)   Eth1/32(P)
cs1#
```

3. Verify that the cluster LIFs have reverted to their home ports:

`network interface show -vserver Cluster`

Show example

```
cluster1::*> network interface show -vserver Cluster
            Logical            Status     Network            Current
Current Is
Vserver     Interface          Admin/Oper Address/Mask       Node
Port    Home
----------- ------------------ ---------- ------------------
------------------- ------- ----
Cluster
            cluster1-01_clus1  up/up      169.254.3.4/23
cluster1-01         e0d     true
            cluster1-01_clus2  up/up      169.254.3.5/23
cluster1-01         e0d     true
            cluster1-02_clus1  up/up      169.254.3.8/23
cluster1-02         e0d     true
            cluster1-02_clus2  up/up      169.254.3.9/23
cluster1-02         e0d     true
            cluster1-03_clus1  up/up      169.254.1.3/23
cluster1-03         e0b     true
            cluster1-03_clus2  up/up      169.254.1.1/23
cluster1-03         e0b     true
            cluster1-04_clus1  up/up      169.254.1.6/23
cluster1-04         e0b     true
            cluster1-04_clus2  up/up      169.254.1.7/23
cluster1-04         e0b     true
cluster1::*>
```

4. Verify that the cluster is healthy:

`cluster show`

Show example

```
cluster1::*> cluster show
Node                 Health  Eligibility   Epsilon
-------------------- ------- ------------- -------
cluster1-01          true    true          false
cluster1-02          true    true          false
cluster1-03          true    true          true
cluster1-04          true    true          false
cluster1::*>
```

5. Verify the connectivity of the remote cluster interfaces:

ONTAP 9.9.1 and later

You can use the `network interface check cluster-connectivity` command to start an accessibility check for cluster connectivity and then display the details:

`network interface check cluster-connectivity start` and `network interface check cluster-connectivity show`

```
cluster1::*> network interface check cluster-connectivity start
```

NOTE: Wait for a number of seconds before running the show command to display the details.

```
cluster1::*> network interface check cluster-connectivity show
                                  Source              Destination
Packet
Node   Date                       LIF                 LIF
Loss
------ -------------------------- -------------------
------------------- -----------
cluster1-01
       3/5/2022 19:21:18 -06:00   cluster1-01_clus2   cluster1-02_clus1
none
       3/5/2022 19:21:20 -06:00   cluster1-01_clus2   cluster1-02_clus2
none

cluster1-02
       3/5/2022 19:21:18 -06:00   cluster1-02_clus2   cluster1-01_clus1
none
       3/5/2022 19:21:20 -06:00   cluster1-02_clus2   cluster1-01_clus2
none
```

All ONTAP releases

For all ONTAP releases, you can also use the `cluster ping-cluster -node <name>` command to check the connectivity:

`cluster ping-cluster -node <name>`

```
cluster1::*> cluster ping-cluster -node local
Host is cluster1-02
Getting addresses from network interface table...
Cluster cluster1-01_clus1 169.254.209.69 cluster1-01     e0a
Cluster cluster1-01_clus2 169.254.49.125 cluster1-01     e0b
Cluster cluster1-02_clus1 169.254.47.194 cluster1-02     e0a
Cluster cluster1-02_clus2 169.254.19.183 cluster1-02     e0b
Local = 169.254.47.194 169.254.19.183
Remote = 169.254.209.69 169.254.49.125
Cluster Vserver Id = 4294967293
Ping status: ......
Basic connectivity succeeds on 4 path(s)
Basic connectivity fails on 0 path(s)
................
Detected 9000 byte MTU on 4 path(s):
    Local 169.254.19.183 to Remote 169.254.209.69
    Local 169.254.19.183 to Remote 169.254.49.125
    Local 169.254.47.194 to Remote 169.254.209.69
    Local 169.254.47.194 to Remote 169.254.49.125
Larger than PMTU communication succeeds on 4 path(s)
RPC status:
2 paths up, 0 paths down (tcp check)
2 paths up, 0 paths down (udp check)
```

What’s next?

After you’ve upgraded your RCF, you verify the SSH configuration.